

Quantum Worlds: coupling, physicality and learned representations

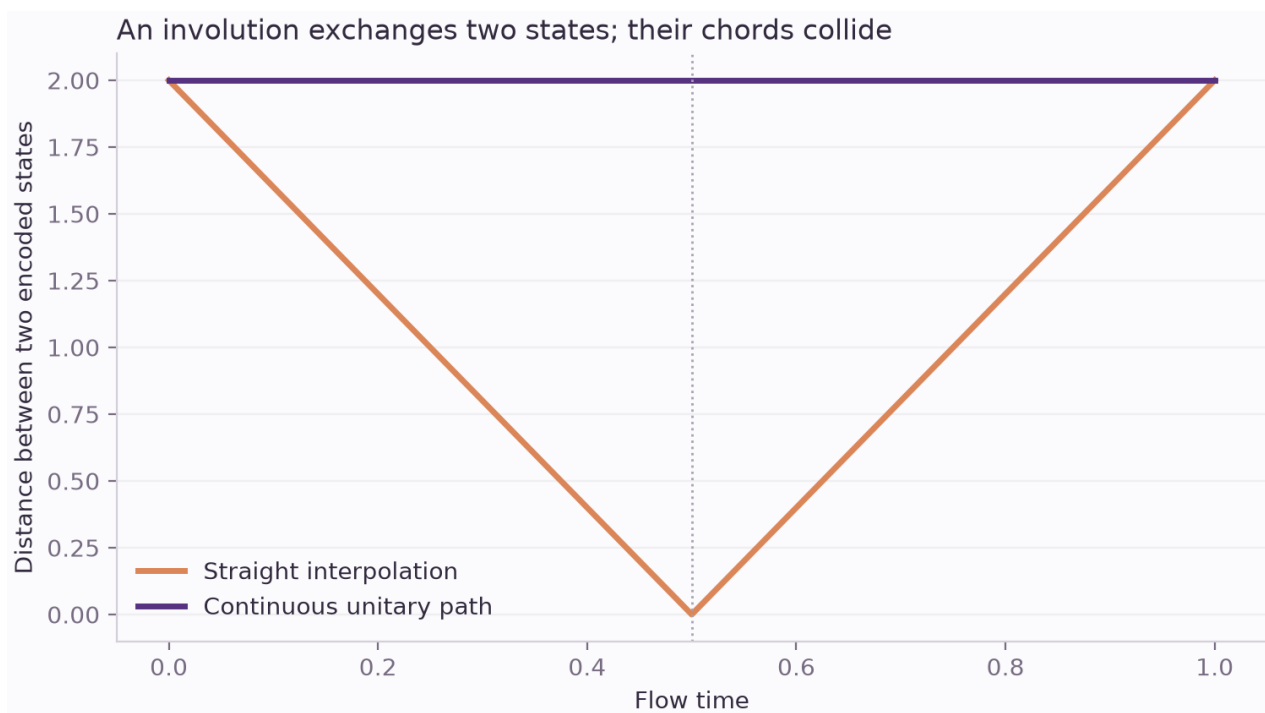
Final report - 3 October 2026. Completed before the 22:00 Toronto deadline.

We built a two-qubit benchmark that separates learning a representation, predicting dynamics, planning controls and identifying noise. Continuous latent generators support accurate 64-step rollouts in eight of ten initial fresh seeds; two failures survive excellent one-step readouts. A later teacher-forced triplet study reaches mean 64-step fidelity 0.9845 and control fidelity 0.9871 across all ten fresh seeds. Its same-encoder classical controller reaches 0.9924. Physics-constrained channel models preserve physical validity, while task-specific measurements improve entanglement-breaking index inference. Readout misspecification can erase that benefit and produce confident errors. All results are local simulations. No quantum computational advantage or breakthrough is established.

1. A concrete flow-matching obstruction

Let an action exchange two distinct states A and B, as an involution can. Their encoded straight paths have the same midpoint, $(E(A)+E(B))/2$, but require opposite velocities $E(B)-E(A)$ and $E(A)-E(B)$. A deterministic field receiving only the intermediate point, time and action cannot supply both. This exact collision holds for any encoder that distinguishes A from B; it is not merely a linear-encoder issue. It identifies an obstruction to fitting these paired paths exactly, not a theorem that every trained model must have poor endpoints.

For H on qubit 0, the Pauli interpolation loses eight of fifteen dimensions at the midpoint. Continuous unitary paths remain distinguishable. Supplying the starting state separately removes this particular ambiguity. More integration steps alone cannot repair missing information.



Midpoint collision in Pauli coordinates

The three-seed controlled-basis screen obtains approximately 0.49 one-step projected fidelity for unconditioned chord flow and 0.93 with source conditioning. Physics-path generator models perform nearly perfectly, but receive derivative labels unavailable to endpoint models. An endpoint-only periodic linear model also performs nearly perfectly with abundant ideal data, preventing a claim that elaborate dynamics are necessary for this simple system.

2. Removing the supervision confound

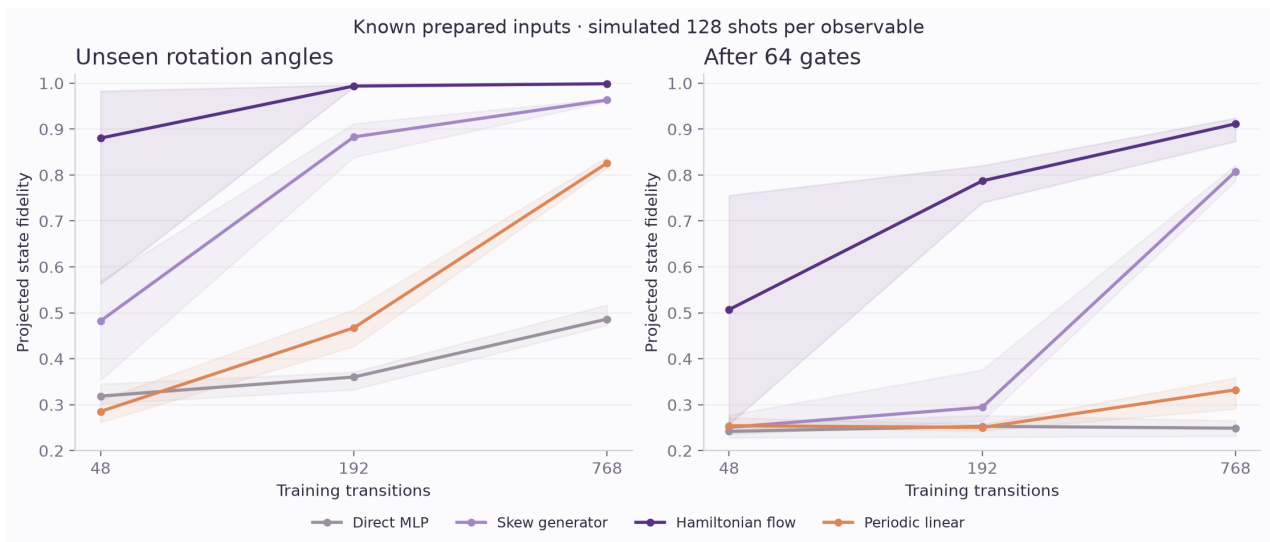
The endpoint-only study gives four model families identical training pairs: direct MLP, skew-symmetric generator, Hamiltonian-commutator generator and periodic linear regression. All use a controlled invertible latent basis. Learning rates or ridge strengths are selected with equal three-choice grids and two independent validation seeds, then frozen before final states are generated. There are 144 validation runs and 120 evaluation runs over six data/noise conditions and five fresh paired seeds.

Training angles lie in $[-1.2, 1.2]$; evaluation uses magnitudes $[1.8, \pi]$. Long rollouts contain 64 unseen action steps. Prepared input states are known exactly; only future Pauli targets receive independent simulated binomial measurement noise. This is a calibration experiment, not a partially observed hardware world model. 128 shots per each of 15 observables means 1,920 simulated shots per target. No hardware was accessed.

At 768 transitions and 128 shots per observable:

Model	Unseen-angle fidelity	64-step fidelity	Raw valid states after rollout
Hamiltonian flow	0.998713	0.911398	100%
Skew generator	0.963284	0.807744	0%
Periodic linear	0.826046	0.332389	0%
Direct MLP	0.486359	0.248885	0.9%

Fidelity is measured after projection to a valid density matrix. Raw validity, Pauli error and correction size are recorded independently, so projection cannot conceal invalid predictions. A skew generator preserves Euclidean norm but need not preserve density-matrix positivity; a Hamiltonian commutator does. With sufficient noiseless data, the linear baseline is again essentially exact.



Endpoint-only sample curves; shading is seed minimum-maximum

The 48-transition setting is not uniformly successful: one noiseless Hamiltonian seed has unseen-angle fidelity 0.792 and 64-step fidelity 0.265. The reported means retain this failure. Five paired seeds limit statistical power: even five wins give a two-sided exact sign-flip p-value of 0.0625. Bootstrap intervals are pointwise, conditional on the study protocol, and not multiplicity-adjusted.

3. Actual nonlinear latent learning

A separate JEPA-style pilot learns an encoder from Pauli observations plus independent nuisance channels. Training uses predictive latent losses and an anti-collapse covariance anchor, with no reconstruction objective. A physical readout is fitted afterward on independent states with the encoder frozen. Readout ceilings distinguish lost representation information from dynamics error. Future information is used for training targets only.

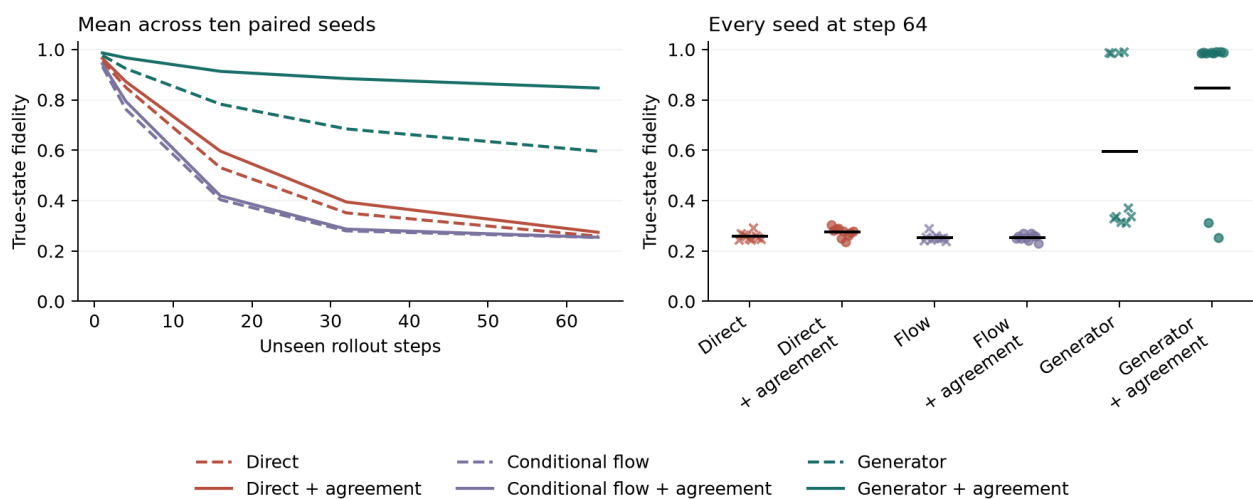
Weak anchoring lets a latent generator collapse to effective rank about one. Stronger anchoring restores effective rank about fourteen and readout fidelity about 0.99. At 10,000 training steps in two validation seeds, a shared latent generator reaches unseen-angle fidelity 0.855-0.872 and 32-step fidelity 0.433-0.441. Matched strong-anchor direct prediction reaches 0.660-0.713 and 0.349-0.390; Gaussian-source conditional flow reaches 0.682-0.698 and 0.290-0.356. These are exploratory results, not a tuned final comparison.

A physical density-matrix latent was also explored. Positivity and Haar second-moment diversity are insufficient by themselves: initial models retain less input information and predict poorly. A linear-encoder/joint-target pilot improves readout fidelity to 0.78-0.81 but still fails long rollouts. These negative results remain part of the evidence. Neither this finite diversity regularizer nor covariance matching certifies a particular distribution.

Fresh confirmation: information is not compositional compatibility

After the pilots, settings were frozen before generating new training and calibration states. Ten fresh initialization/evaluation seeds compare three families with and without same-state nuisance-view agreement: 60 runs, each with 4,000 training steps, covariance-anchor weight 20 and a frozen encoder. Independent evaluation readouts output positive, unit-trace density matrices; their mean fidelity ceilings are approximately 0.999 in all conditions. The readout receives calibration labels only after encoder training and cannot improve the encoder. The world-model training contains no reconstruction loss.

Model	View agreement	Unseen-angle fidelity	64-step fidelity	Seeds above 0.95 at 64 steps
Direct predictor	No	0.7361	0.2585	0/10
Direct predictor	Yes	0.7807	0.2741	0/10
Conditional Gaussian flow	No	0.6398	0.2535	0/10
Conditional Gaussian flow	Yes	0.6971	0.2543	0/10
Continuous generator	No	0.9588	0.5957	4/10
Continuous generator	Yes	0.9766	0.8469	8/10



Fresh confirmation, including every seed

The generator is trained by integrating its action-conditioned linear latent field and matching future endpoints. It is **not** trained with a flow-matching velocity objective. The Gaussian conditional-flow comparator receives the initial encoding as a condition and a fresh Gaussian source at every inference step. It forecasts one step autoregressively, rather than reproducing Flow-JEPA's full future-trajectory generator. Architecture settings were pilot-selected; this comparison does not establish the best achievable version of each family.

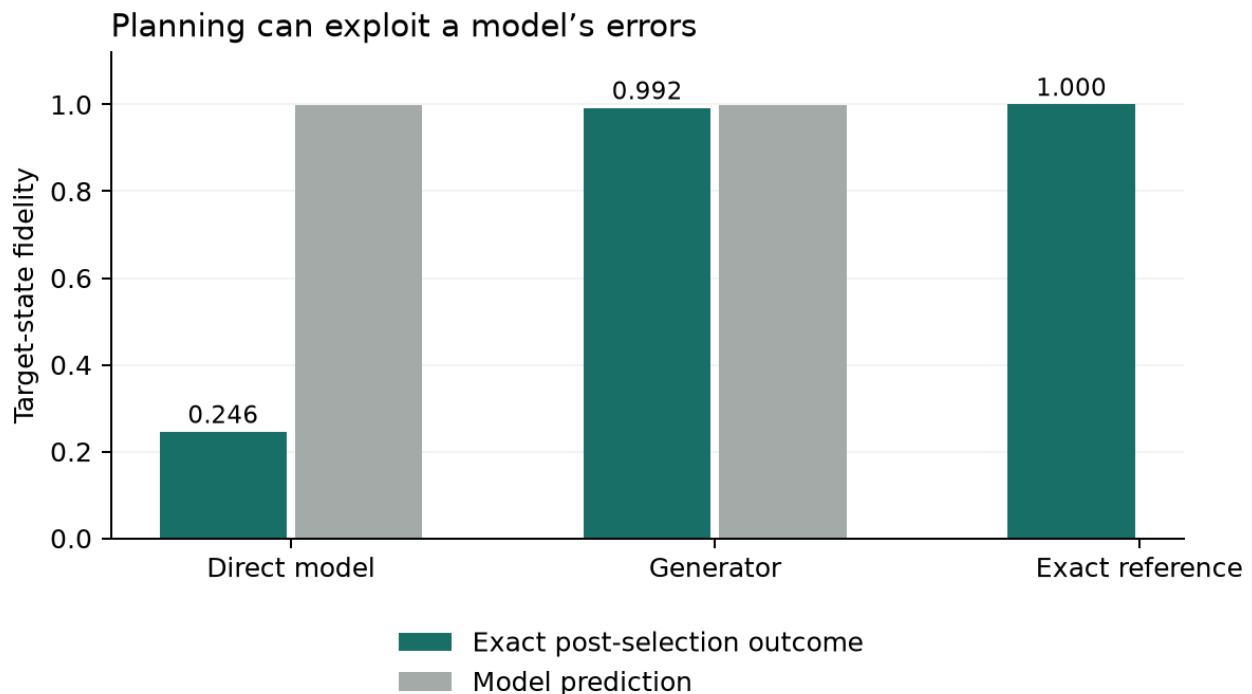
With view agreement, generator-minus-direct paired differences are 0.1959 for unseen angles and 0.5728 for 64-step fidelity. Exact paired sign-flip tests give Holm-adjusted $p=0.0195$ and 0.0234 across ten declared contrasts. Corresponding pointwise bootstrap 95% intervals are $[0.1658, 0.2227]$ and $[0.3627, 0.7191]$. These intervals concern seeds conditional on the protocol, not all datasets or architectures. The generator's own invariance improvement is not significant after that adjustment (angle $p=0.5664$; rollout $p=0.4141$); its apparent pilot benefit was too strong a causal interpretation.

Successful generators learn nearly affine Pauli coordinates: relative squared linear-fit error is about 0.003 on independently sampled states. Failing seeds retain state information but learn less compatible nonlinear coordinates and larger involution/commutation residuals. This is an association, not yet a causal proof. Exact same-generator composition and inversion are built into the exponential; cross-gate relations and physical decoded outcomes are separate checks. Gaussianity is unnecessary for these state coordinates.

Frozen-model control and model exploitation

A preselected smoke checkpoint plans continuous RZ angles in a fixed 43-gate H/CX template: four local Euler layers, three alternating CNOTs and 24 adjustable angles. On eight fresh initial states, four Bell targets and four independent Haar targets, each model receives 250 Adam steps and four identical random restarts. Selection uses only learned latent goal distance. Exact simulation evaluates the selected controls afterward and never selects a restart.

Planner model	True target fidelity	Bell targets	Arbitrary targets	Predicted target fidelity
Frozen direct model	0.2456	0.1862	0.3049	0.9983
Frozen continuous generator	0.9921	0.9934	0.9909	0.9991
Exact Pauli reference	Approximately 1	Approximately 1	Approximately 1	-

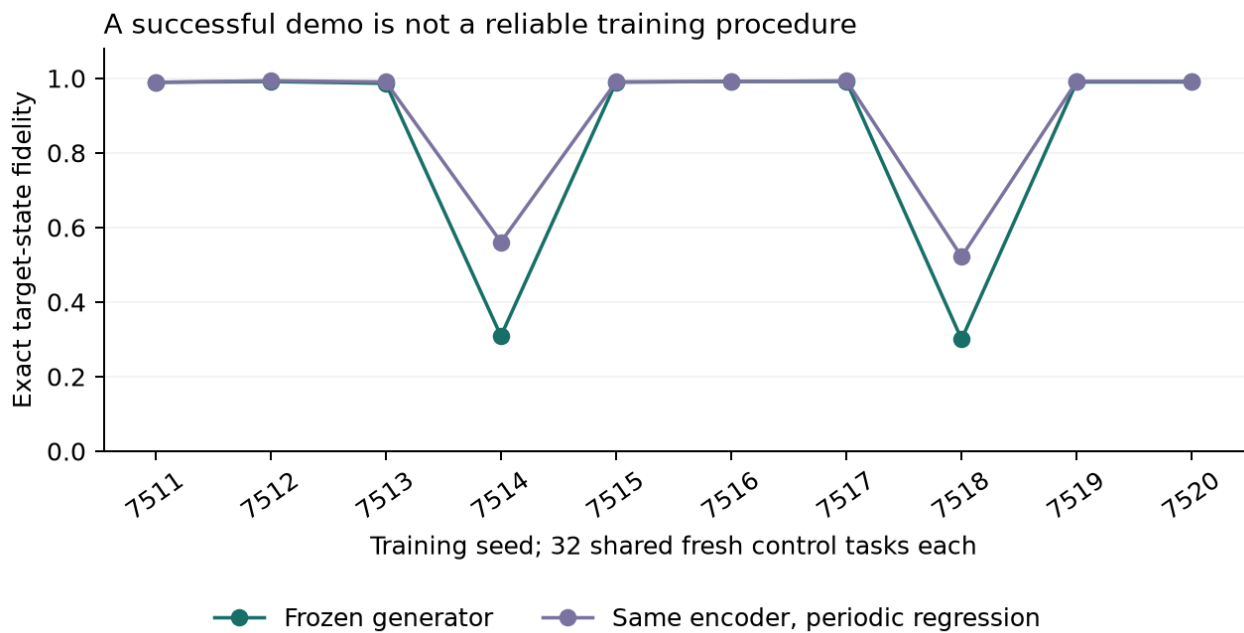


Predicted versus actual control outcomes

The direct model's self-assessment is misleading: optimizing against it finds circuits that exploit prediction error. A low internal goal loss is therefore insufficient evidence of control success. The generator supports a useful local control demo, but these eight tasks use one training seed and do not erase the failures in the independent ten-seed study. Exact two-qubit control is inexpensive and established; this is utility evidence, not quantum advantage.

The all-seed follow-up reproduces every frozen generator checkpoint with exactly zero discrepancy in saved rollout fidelities, then plans for 32 fresh shared tasks per seed. Eight seeds achieve mean target fidelity 0.986-0.992; two fail at 0.300 and 0.309. The across-seed mean is 0.8529. Transferred periodic regression on the same generator encoders averages 0.9013, but still fails at 0.522 and 0.561 on the two problematic encoders. On the first predefined direct-model checkpoint, the same classical transfer improves planning from 0.3352 to 0.9686. The exact reference remains essentially perfect. The two failing generators forecast target fidelity 0.882 and 0.970 despite actual values near 0.3. These fresh

control tasks confirm utility and expose reliability limits; retraining the same seeds is reproduction, not additional independent evidence.



Planning across every frozen training seed

Strong classical predictor: much of the deficit is in dynamics

On the two first, preselected saved checkpoint seeds, periodic affine regression is fitted to the same training endpoints after freezing each encoder. It uses constant discrete-gate maps and the basis $[1, \cos(\theta), \sin(\theta)]$ for RZ gates. This is a transferred-encoder comparator, not an independently trained encoder. All models then face the same fresh evaluation states and the same readout.

Direct encoders support 64-step fidelity 0.805-0.950 with the linear predictor, versus 0.250-0.285 with their original MLP. Conditional-flow encoders support 0.960-0.969, versus 0.239-0.268 originally. Generator encodings support 0.987-0.989, approximately matching their original generator. Thus, much of the original gap is a predictor/inductive-bias limitation rather than unusable representations. This limited checkpoint comparison reinforces the need for a strong classical baseline and weakens any claim that flow integration itself is indispensable. Raw Pauli inputs already make this small system tractable.

A first four-seed algebra-prior intervention is negative: imposing strong operator-level involution and RZ-commutation penalties from initialization lowers readout ceilings to 0.907-0.928 and 64-step fidelity to 0.247-0.256, compared with approximately 0.988 for its paired unregularized generators. Correct algebra alone does not ensure informative coordinates or successful optimization. This intervention is exploratory and separate from the frozen ten-seed confirmation.

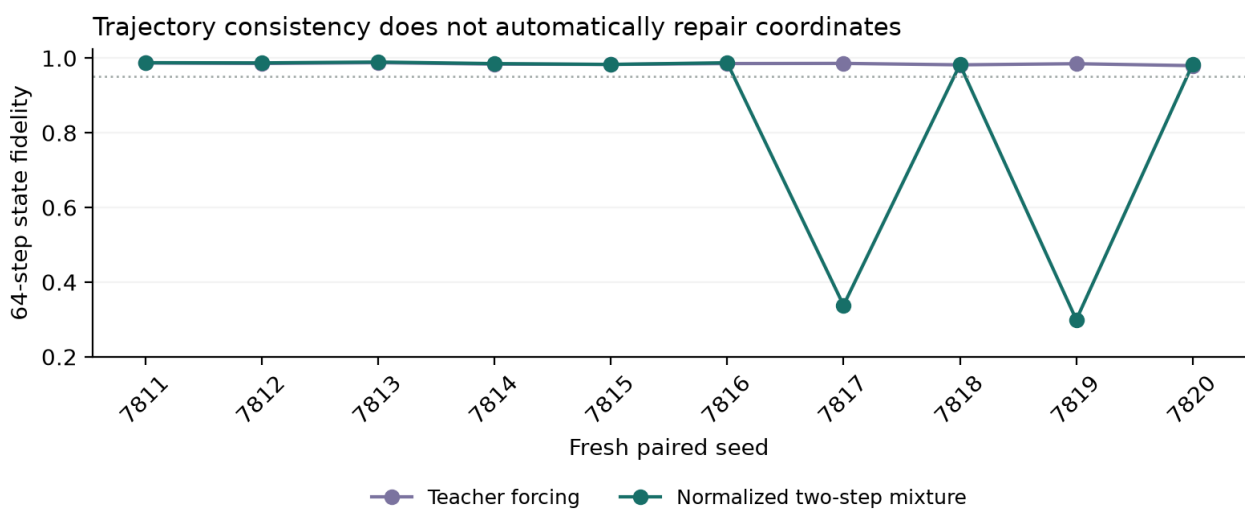
Short-trajectory training: a failed intervention and a loss-scale confound

The final exploitation screen compares the same 4,096 observed two-action triplets across four fresh seeds. Both conditions use intermediate and final encoder targets, the same nuisance views, and the same covariance anchors. Adding a two-step free-running endpoint loss consumes more compute but no extra observed labels. Direct predictions remain poor under either objective. Generator one-step training reaches 64-step fidelity 0.980-0.984, while adding the equally weighted rollout term yields 0.223-0.250 and degrades readout information. This particular intervention fails in all four generator seeds.

However, addition doubles total predictive weight relative to regularization. A diagnostic normalizes the teacher/rollout mixture by its total weight. Three seeds recover to 0.981-0.987, while one still fails at 0.307 despite a readout ceiling near 0.999. This separates a loss-balance problem from the remaining composition problem; it does not prove that trajectory consistency is harmful in general or that normalization improves reliability. A fixed-setting fresh paired diagnostic was queued while the fourth result arrived; the screen did not pass the proposed positive advancement gate. We retain that chronology rather than labeling it a selected success.

The fresh diagnostic uses new training/calibration states and ten new seeds:

Generator training	Unseen-angle fidelity	64-step fidelity	Seeds above 0.95
Teacher-forced triplets	0.9937	0.9845	10/10
Normalized one-step/two-step mixture	0.9813	0.8524	8/10



All fresh triplet-study seeds

Seven paired seeds show tiny rollout gains for the mixture, but two fail severely. The mean paired difference is -0.1322; its pointwise bootstrap 95% interval is [-0.3308,0.0018] and exact two-sided sign-flip $p=0.4980$. Ten seeds do not establish a significant mean effect. They do refute a claim that this fixed intervention removes composition failures. Readout ceilings remain near 0.999 in both conditions. Compared with the earlier single-transition study, teacher forcing also changes training data and anchor averaging, so its apparent reliability improvement cannot be attributed causally to one component.

Every frozen checkpoint then plans on the same 32 new Bell/Haar target tasks. This produces 41 comparisons: twenty models, twenty same-encoder periodic-linear predictors fitted to their saved training transitions, and an exact reference. Latent cost selects controls; exact outcomes are evaluated afterward. Teacher forcing gives mean control fidelity 0.9871, with all ten seed means between 0.9842 and 0.9912. The classical transfer gives 0.9924. The normalized mixture averages 0.8636, or 0.8972 with classical transfer; both preserve its two failed encoders. The exact reference is essentially perfect. This is a useful local control demo with strong classical competition, not a quantum speedup. Seeds share one training dataset and one task set within this study; broader data and observation-model reliability remains untested.

The teacher-forced models also create the specified Bell state with mean fidelity 0.9920 and negativity 0.4948 (ideal: 0.5). Their decoded predictions for H/RZ in each order have fidelity about 0.9942. Mean

latent residuals for H and CX involutions and independent-qubit RZ commutation are 0.00043, 0.00041 and 0.00048. RZ inversion and same-generator angle addition are exact to numerical precision because the exponential builds them in; they are not evidence that the model discovered those identities. These checks show usable relational and compositional predictions in the chosen complete-state setting.

4. Physical contraction is not representation collapse

The noise extension calibrates global two-qubit depolarization and amplitude damping on qubit 0. It uses known physical Pauli coordinates, a known dictionary of channel types, and learns only their two positive rates in the physical model. Skew and unconstrained affine generators are larger baselines. This is a structural boundary experiment, separate from nonlinear JEPA: no claim of learned perception, unknown-noise discovery, or hardware validation follows.

There are 36 exploratory runs: three paired seeds, 128 or 512 transitions, noiseless or 128-shot output observables, and three models. Times are trained in [0.05,0.8] and evaluated in [1.2,2.0]. Inputs are known prepared density matrices; all models receive the same noisy endpoint labels. Negative noise times are forbidden. At 512 transitions and 128 shots per observable:

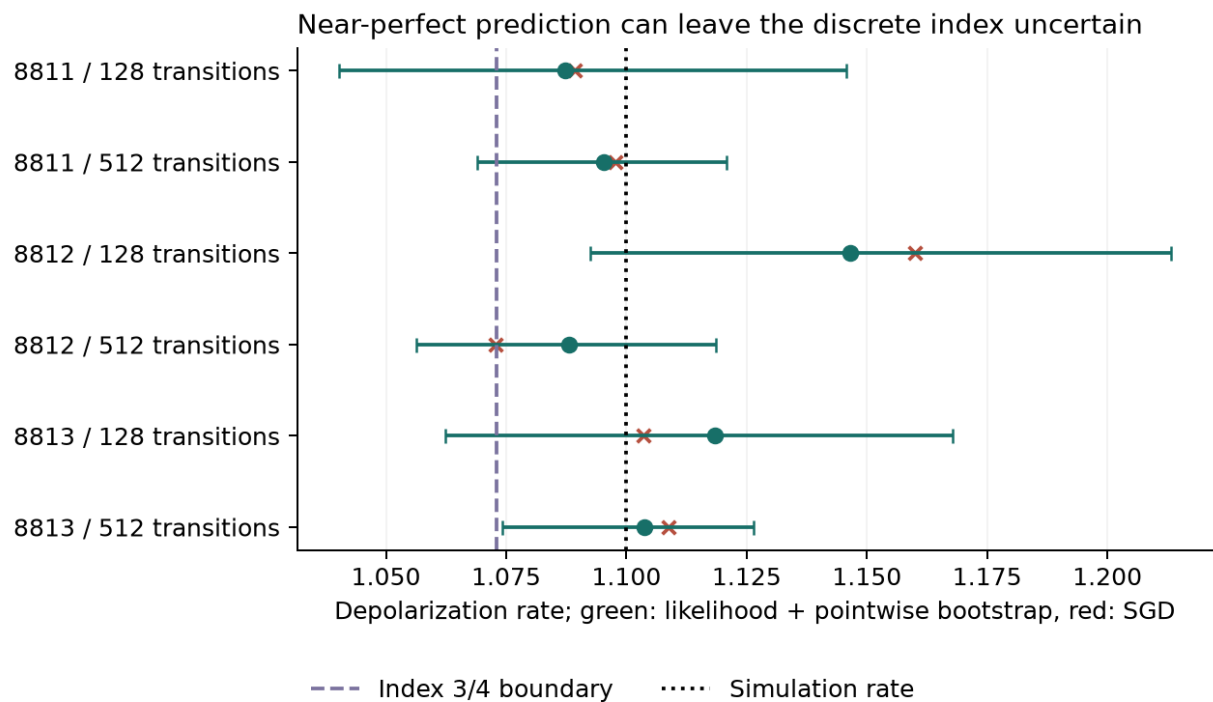
Model	Unseen-time fidelity	32-step fidelity	Raw valid states at unseen times
Physical channel dictionary	0.999983	0.999984	100%
Skew generator	0.733414	0.589951	23.4%
Affine generator	0.983478	0.996079	72.9%

Fidelity here is squared Uhlmann fidelity for mixed states, after density repair where needed. A maximally mixed predictor already scores 0.9464 at step 32: noise makes late prediction easier by forgetting initial states. Long-rollout fidelity alone is therefore insufficient. Complete positivity is checked with normalized Choi matrices, separately from positivity on sampled states: the physical model passes 96/96 sampled gate/time checks, the skew model 0/96 and affine models 35/96. The physical dictionary is CPTP by construction for positive rates and time; sampled checks alone do not prove a general model CPTP.

For global depolarization, centered linear encodings obey $z_{\text{future}} = p z_{\text{current}}$. If current covariance is sl , future covariance is $p^2 sl$. Anchoring both to I has per-direction objective $[(s-1)^2 + (p^2 s - 1)^2]/2$, minimized at $s = (1+p^2)/(1+p)$, with a strictly positive residual whenever $0 < p < 1$. The implementation's mean over all covariance entries adds a factor $1/D$ to this isotropic expression. Thus, source/future whitening conflicts with genuine contraction for a linear informative encoding. This is a concrete consistency check, not a theorem that every nonlinear encoder must fail. Artificial nuisance invariance and real physical loss of distinguishability need different treatment; future isotropy should not be imposed blindly.

The same channel links prediction to entanglement loss. For $d=4$ depolarization, $p(t) = \exp(-\gamma t)$, and the known isotropic-channel threshold is $p=1/5$: $t_{\text{EB}} = \log(5)/\gamma$. For fixed step Δ , its entanglement-breaking index is $\text{ceil}(\log(5)/(\gamma \Delta))$. This follows from the established [depolarizing-semigroup analysis](#). Our simulated $\gamma=1.10$ gives $t_{\text{EB}}=1.4631$ and index three at $\Delta=0.5$. At that threshold, worst-case trace distance from the stationary state remains 0.15: entanglement breaking is not complete input forgetting. Choi partial transpose eigenvalues cross zero as expected; sufficiency uses the isotropic family, not a generic 4×4 PPT test. These are known formulas validated by the experiment, not new quantum theory.

One noisy stochastic-optimizer estimate gives an index of four despite unseen-time fidelity 0.999976. A discrete threshold can be wrong when continuous prediction looks excellent. Model-conditional point estimates are not uncertainty certificates for the true channel. A separate classical full-batch likelihood fit corrects that point estimate to index three. However, 200 parametric bootstrap replicates for each of the six noisy fits yield pointwise intervals that span indices three and four in four cases. These are approximate intervals conditional on the known channel dictionary, prepared inputs and independent binomial measurements; they provide no simultaneous-coverage or hardware-model guarantee. Accuracy, optimizer convergence, model assumptions and uncertainty must all be checked before interpreting a predicted entanglement-loss index.



Rate uncertainty relative to the discrete-index boundary

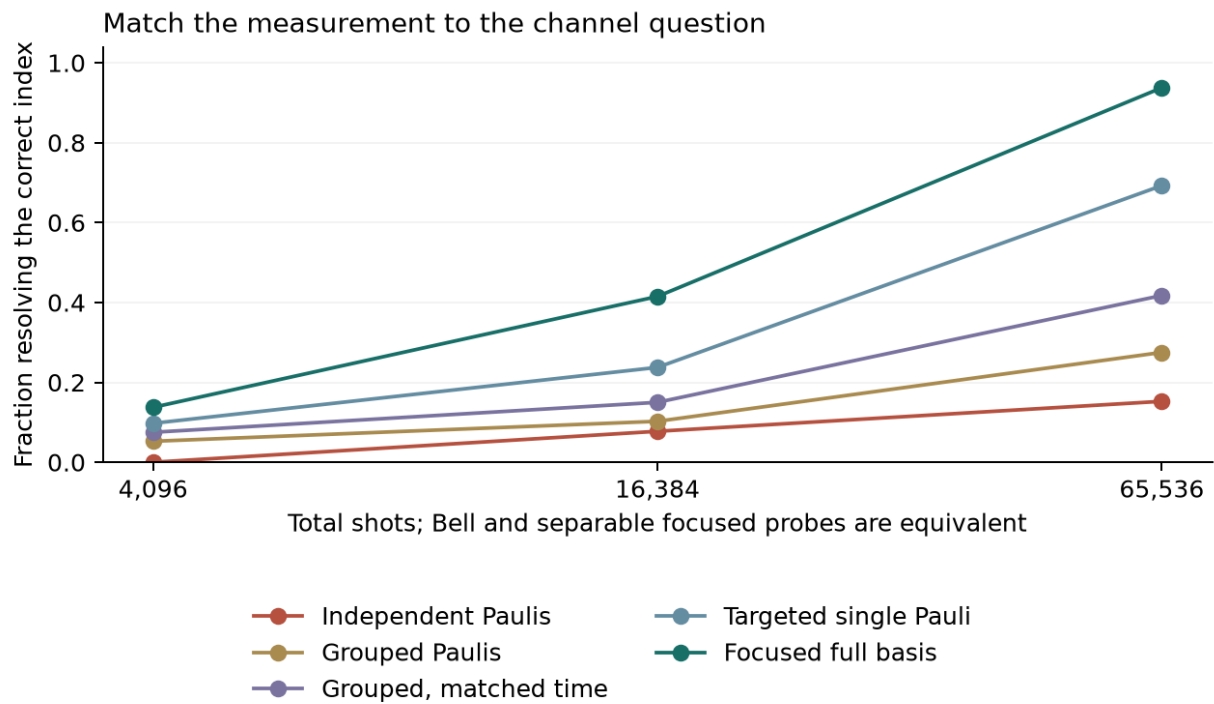
5. Measure the channel question, not every observable

The uncertain index motivates a separate, inexpensive measurement experiment. For the known four-dimensional isotropic depolarizing family, prepare $|00\rangle$, evolve for a fixed dimensionless time $t=0.8$, and measure both qubits in Z . The probability of returning 00 is $q=(1+3 \exp(-\gamma t))/4$. The complete basis outcome therefore estimates the channel rate directly; marginalizing to one qubit discards information. An exact binomial Clopper-Pearson interval for q maps monotonically to a rate interval and then to a set of possible indices. We call an index resolved only when that entire interval implies one integer. This is conditional on the specified channel family, preparation and readout. It is not a general channel certificate.

We froze six designs after two cheap screens, then ran 400 fresh measurement replicates per design and shot budget. Tomography uses 64 independently prepared Haar states. Grouped tomography obtains three compatible Pauli observables from each joint measurement, avoiding a straw-man comparison to ungrouped shots. The matched-time variant uses $t=0.8$ for every tomography input, isolating the advantage from an unnecessarily short exposure. Total shots are equal; state preparation and circuit overhead are not included in this resource comparison. Tomography intervals use an approximate profile-likelihood construction; the targeted designs use exact binomial intervals. Their different interval methods must be considered alongside the independently reported rate RMSE.

At 65,536 total shots and $\gamma=1.10$, the true step-0.5 index is three:

Measurement design	Correctly resolved index	Rate RMSE	Empirical interval coverage
Independent Pauli measurements	15.25%	0.03234	95.00%
Grouped Pauli measurements	27.50%	0.01899	94.75%
Grouped, matched exposure time	41.75%	0.01513	96.75%
Targeted single-qubit marginal	69.25%	0.01112	94.50%
Focused full-basis measurement	93.75%	0.00783	94.25%



Equal-shot measurement comparison

The gain is task-specific measurement design. A Bell preparation followed by inverse Bell measurement has exactly the same probabilities as this product preparation under global isotropic noise. We verify the equivalence against Qiskit and reuse the same counts for this analytic control; it is not an independent experiment or evidence of entanglement advantage. The single-qubit marginal is likewise derived from the focused counts. Budgets share random seeds, so neither budgets nor these controls may be pooled as independent data. All 7,200 comparisons are preserved, with 400 independent replicates within a design/budget. Pointwise Monte Carlo proportion intervals are saved separately. Nominal exact coverage under the model does not require a finite Monte Carlo coverage fraction to equal 95%.

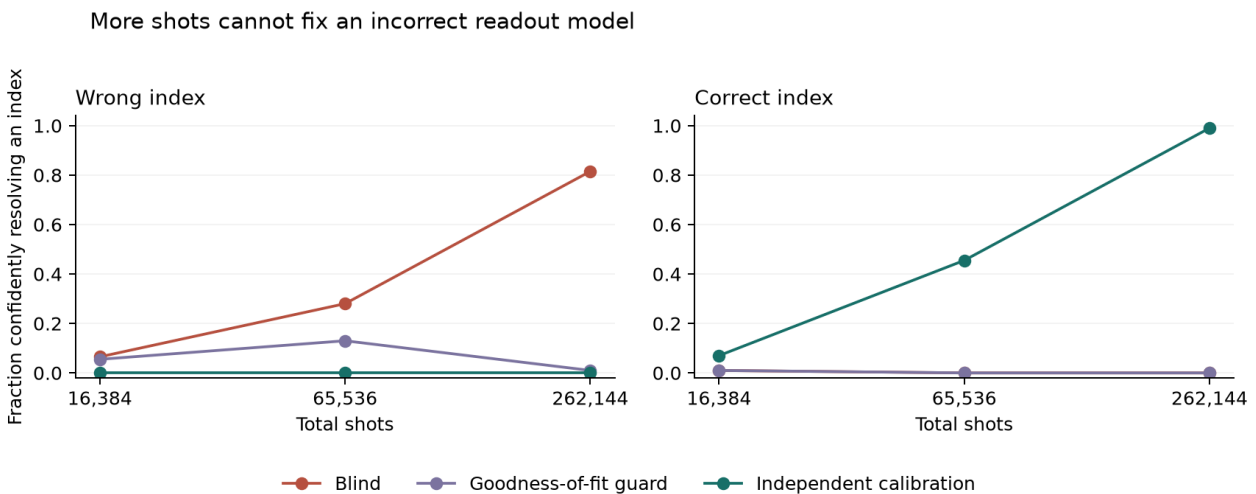
More shots can make an incorrect model more confidently wrong

We challenged that result with independent symmetric 1% bit-flip readout error on each qubit. For $\gamma=1.05$, the true step-0.5 index is four. Ignoring readout error biases the inferred decay rate upward toward index three. A goodness-of-fit guard tests the unequal nonreturn outcomes and withholds a result when its approximate Pearson test rejects the ideal model. A calibrated estimator reserves one quarter of the same total budget for independent prepared- $|00\rangle$ readout references and uses the remaining three quarters for the channel probe.

Two exact 97.5% binomial intervals, one for readout error and one for return probability, give a conservative joint rate interval by Bonferroni and monotonic propagation. The guarantee still assumes identical independent bit-flip errors, ideal reference preparation, and unchanged readout between reference and probe. We do not know how it performs under arbitrary asymmetric or correlated errors.

At 262,144 total shots, across 200 fresh trials:

Estimator	Correctly resolved	Confidently wrong	Withheld	Underlying rate interval coverage
Blind ideal-readout model	0%	81.5%	0%	0%
Blind model with fit guard	0%	1.0%	99.0%	0%
Independently calibrated	99.0%	0%	0%	99.5%



Readout misspecification and calibration

The fit guard is useful but is not a certificate: its two accepted trials are wrong, and the quoted underlying interval coverage is not selective coverage conditional on acceptance. Calibration has a cost under ideal readout: at 65,536 shots, correct resolution falls from 93.5% to 77.5% because reference shots and conservative uncertainty propagation consume budget. This stress study contains 3,600 policy comparisons, with 200 independent replicates within each scenario/budget; guarded results reuse blind observations. Rates, counts, shot accounting and all shared-control dependencies are audited. Model checks and calibration matter more than simply increasing shots near a discrete index boundary.

6. What is and is not new

[Flow-JEPA](#) already combines latent prediction with flow matching and retains the initial observation as a condition; our unconditioned chord witness does not refute it. [Semigroup-JEPA](#) already studies physical generalization. [Quantum Flow Matching](#) already studies density-matrix generation with quantum circuits, and [Hamiltonian learning](#) is established.

Our defensible output is an audited benchmark, a quantum-gate illustration of paired-path ambiguity, and controlled evidence separating norm preservation from physical validity. The midpoint argument is an elementary application of trajectory-crossing ambiguity; novelty has not been established. Successful Hamiltonian calibration is useful, but does not establish a new algorithm. Rotation composition guaranteed by an exponential is an architectural prior, not a discovered law. Learned commutation, noncommutation and Bell-state predictions are evaluated separately in the saved metrics.

The measurement result is a useful channel-identification demonstration, not a new quantum algorithm. The finite-shot confidence construction uses established binomial inference. The combined benchmark exposes three practical failure modes: incompatible learned coordinates, invalid unconstrained channel forecasts, and confident index errors under readout misspecification. The evidence supports these scoped observations rather than a claim of a new general world model.

7. Evidence and recommended direction

- artifacts/quantum-world-screen-v2-20261003: controlled-basis, three-seed screen.
- artifacts/quantum-world-endpoint-confirmation-20261003: frozen selections, paired data hashes, all runs, checkpoints and completion counts.
- artifacts/quantum-world-jepa-screen-v2-20261003, quantum-world-jepa-strong-anchor-20261003 and quantum-world-jepa-long-generator-20261003: nonlinear encoder pilots.
- artifacts/quantum-world-density-*: exploratory physical-latent pilots.
- artifacts/quantum-world-jepa-confirmation-20261003: 60 fresh paired runs, positive readouts, gate-algebra diagnostics and multiplicity-adjusted tests.
- artifacts/quantum-world-control-smoke-20261003: frozen checkpoint hashes, all task states, selected controls and post-selection exact outcomes.
- artifacts/quantum-world-transferred-linear-20261003: same-encoder strong classical predictors, with independent paired evaluation states.
- artifacts/quantum-world-algebra-exploration-20261003: all eight runs of the failed strong-algebra-prior intervention.
- artifacts/quantum-world-noise-channels-20261003: 36 endpoint fits, mixed-state metrics, structural Choi audit, and conditional entanglement-breaking times.
- artifacts/quantum-world-control-confirmation-20261003: all ten reproduced checkpoints and 23 planning comparisons on fresh shared tasks.
- artifacts/quantum-world-noise-rate-audit-v2-20261003: classical likelihood fits and 1,200 parametric-bootstrap fits. The first serialization failure is preserved separately and contains no scientific results.

- artifacts/quantum-world-measurement-confirmation-20261003: frozen six-design equal-shot confirmation, raw counts, shared-control and seed accounting.
- artifacts/quantum-world-readout-stress-v2-20261003: ideal and misspecified readout scenarios, calibration uncertainty and failed-model guard. The first implementation failure occurred before any scientific results and is retained.
- artifacts/quantum-world-sequence-screen-20261003: all 16 matched-triplet teacher-forced/short-rollout training comparisons, including failed generators.
- artifacts/quantum-world-sequence-normalized-20261003: eight same-seed diagnostic runs isolating total predictive loss scale, including the failure.
- artifacts/quantum-world-sequence-confirmation-20261003: twenty fresh paired runs with the advancement-decision chronology preserved.
- artifacts/quantum-world-sequence-control-20261003: 41 frozen-checkpoint planning comparisons, task states, controls and exact outcome reproduction.
- artifacts/quantum-world-sprint-20261003: regenerated figures and paired statistics.

The feasible project is **physics-informed quantum dynamics with audited control and channel uncertainty**. Use the teacher-forced latent generator as the learned unitary world-model demonstration, retain the same-encoder classical regression baseline, and show physical channel identification and calibrated index inference as a separate noisy-system experiment. We have not integrated a nonlinear noisy JEPA model end to end. Full Pauli observations are informationally complete and the 15-dimensional latent is not compression or image perception. Exact statevectors provide reference outcomes; a learned raw-statevector comparator was not included. Training targets use the current encoder with stopped gradients, rather than a separate EMA target encoder. Exact NumPy/SciPy evolution generates states and Qiskit independently checks conventions and channels. Random long compositions are evaluated; a special held-out composition grammar is not defined. Useful next work would improve seed reliability and test genuinely partial observations before adding a larger encoder or another quantum regularizer.

The research cycle balanced cheap mechanism screens with independent confirmation and adversarial checks. Failed density latents, strong algebra penalties, source-free interpolation and short-rollout training are retained. The ledger records decisions and wall-clock progress; manifests bind protocols, source snapshots and report figures. All 149 tests and the FlyWalk smoke check pass. No hardware evidence is claimed.

Reproduce frozen study figures with `uv run python -m scripts.quantum_world_report`. See `PROTOCOL.md` and the artifact manifests for experiment settings. The original screen with a numerical projection failure and the first interrupted JEPA pilot are preserved with explicit validity/failure notes; they are not final evidence.

Regenerate the measurement and trajectory summaries with `uv run python -m scripts.quantum_world_measurement_summary` and `uv run python -m scripts.quantum_world_sequence_summary`. Build the PDF from this source with `uv run --with reportlab==4.4.9 python -m scripts.quantum_world_pdf`. The final completion audit binds evidence, figures, report source and PDF hashes.